Weak isogeny classes: GHS descent and Smith's genus-3 isogeny

Run report `run-20261013T0342Z` · study `research/weak_isogeny_classes_20261013` · 2026-10-13

Run. `cargo run --release --example weak_class_coverage` (code at commit `f26274dda`. The manifests' `git_head` `ee7d9ce1e` is read when the run ends. It is a data-only commit on top of `f26274dda`, so the code is identical). GHS arm: `--ghs 8,9,10,12,14,15,16,18,20,21,22,24 --walk-max 22`, 8 min 38 s. Smith arm: `--smith 3,5,7,11`, 9 min 40 s (rayon, 4 threads). Host: 4-vCPU Intel Xeon @ 2.10 GHz cloud container with no isolation. Timings are incidental. Both arms completed with no failed runs. Raw records: `data/ghs_N*.json`, `data/smith_p*.json`, `data/smith_chebotarev.json`, `data/run_manifest_*.json`. All counts are **measured** by exhaustive enumeration; "derived" marks a consequence of the counts plus a cited theorem. Seeds: none (deterministic enumeration).

1. Question

The DLP transfers along an explicit isogeny, so the weak set is a union of weak isogeny classes. (1) GHS: what fraction of binary curves $E_b : y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^N}$, $N = nl$, lies in a trace class (Tate: $\mathbb{F}_{2^N}$-isogeny class) containing a curve with $2 \le m(b) \le M$? How much of that do 3-isogeny walks reach? (2) Smith: what fraction of genus-3 curves $y^2 = f(x)$, $\deg f = 8$, over $\mathbb{F}_p$ has a rational tractable $(\mathbb{Z}/2)^3$ kernel? What does closing under the isogeny class add?

2. Definitions (checked against sources)

- Magic number: $m(b) = \dim_{\mathbb{F}_2} \mathrm{Span}\{(1, \sqrt{b_i})\}$ (Maurer–Menezes–Teske 2002, eq. (5)). The repository's older `ec_trapdoor::magic_number` drops the leading 1. It undercounts by one exactly when $(x-1) \nmid \mathrm{Ord}_b$ (MMT Thm 5(i)); test `plain_orbit_undercounts`.
- Twist (Tr $a = 1$) is admitted only under MMT Lemma 6 (8): n odd or $\mathrm{Tr}_{K/k}(b) \neq 0$. Subfield curves ($m = 1$) are never weak sources.
- Traces come from Kloosterman sums via one Walsh–Hadamard transform: $t(b) = -W(\texttt{sqrt}\ b)$. They are checked against naive counts.
- 3-walk: roots of $\Phi_3(X, j) \bmod 2$ plus Frobenius $j \mapsto j^2$ (free); 0-1 BFS from the weak vertices.
- Smith tractable subgroup (Smith 2009, Def. 1): a Galois-stable partition of the 8 Weierstrass points into 4 pairs. Classes are L-polynomials from exact counts over $\mathbb{F}_p, \mathbb{F}_{p^2}, \mathbb{F}_{p^3}$.

3. Lead 1: GHS weak set per curve and per class

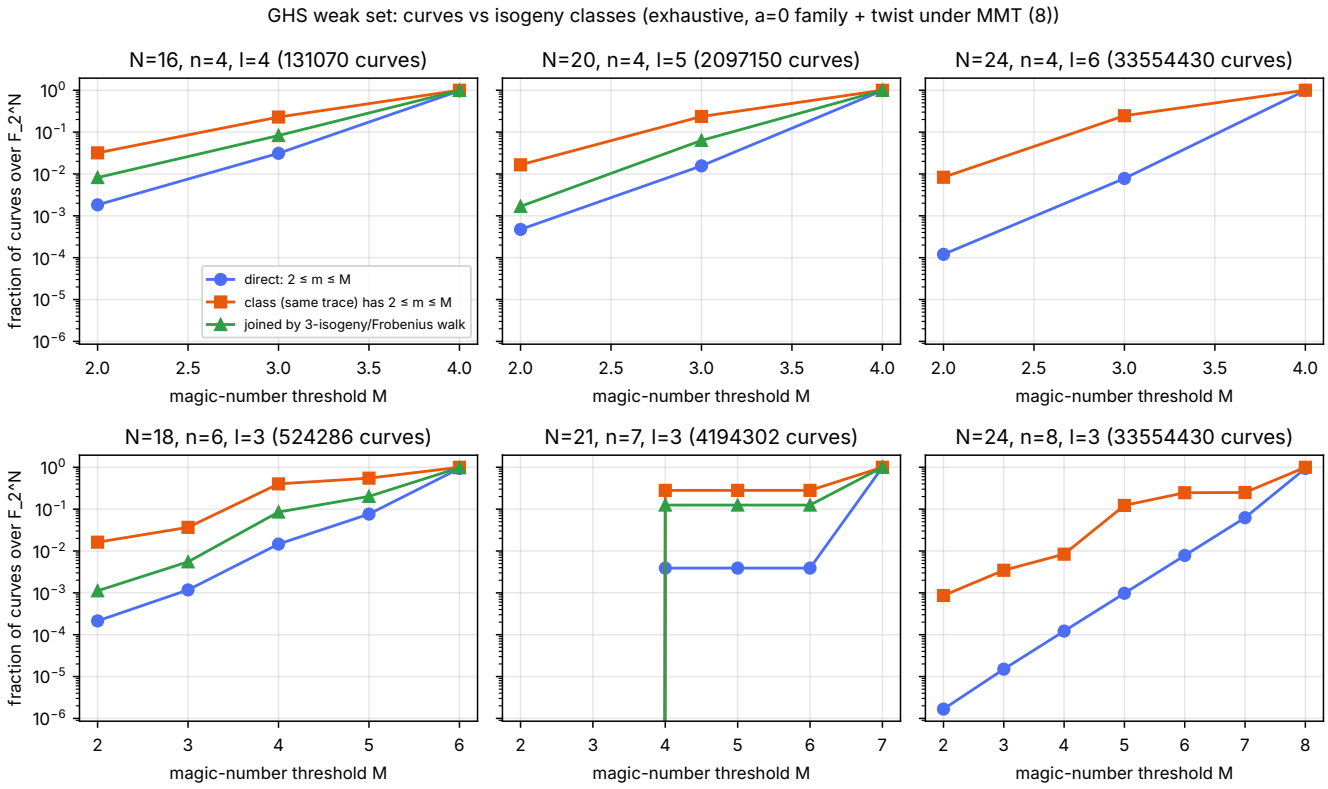


Figure 1: Fraction of all $2(2^N - 1)$ ordinary curves over $\mathbb{F}_{2^N}$, exhaustive: directly weak, class-covered and 3-walk-covered, against the threshold M. Zeros drop off the log axis. 3-walk built for $N \le 22$.

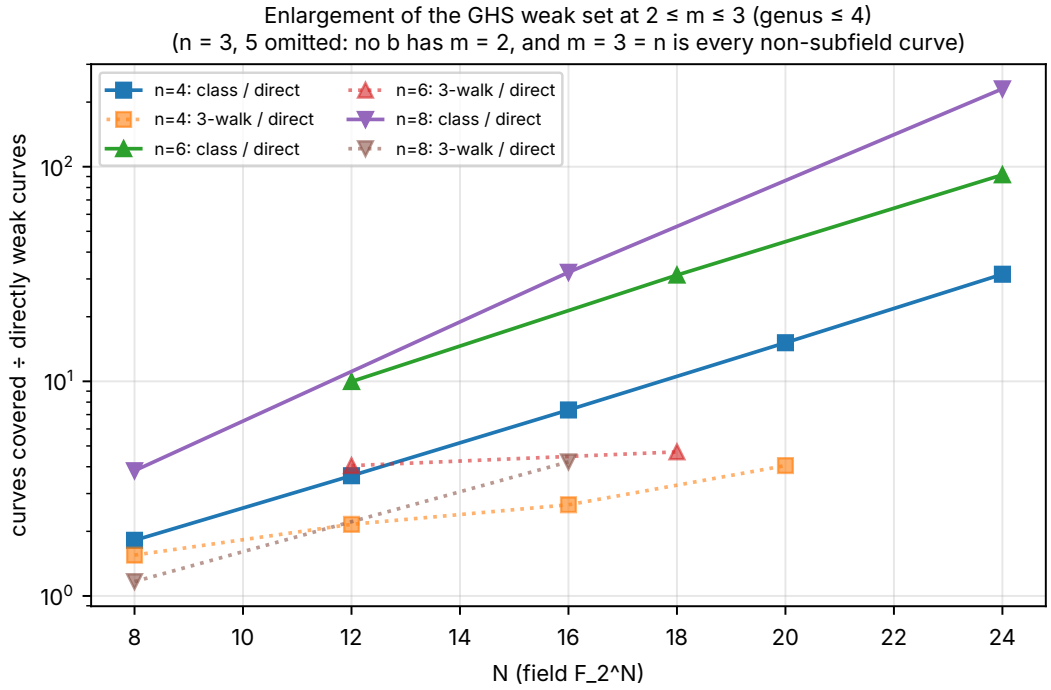


Figure 2: Class-covered ÷ directly weak curves at $2 \le m \le 3$ by N. Dotted lines: the same ratio for 3-walk-covered curves.

N	n	l	M	curves	classes	direct	class	3-walk	class÷direct	weak classes	median log2 sqrt(class)	max walk steps
8	2	4	2	510	32	9.41e-01	9.80e-01	9.57e-01	1.0	30	2.00	1
8	4	2	2	510	32	2.35e-02	8.82e-02	5.69e-02	3.8	3	2.16	1
8	4	2	3	510	32	1.18e-01	2.14e-01	1.82e-01	1.8	7	2.00	1
10	2	5	2	2046	64	9.70e-01	9.95e-01	9.84e-01	1.0	62	2.45	1
12	2	6	2	8190	128	9.85e-01	9.97e-01	9.89e-01	1.0	126	2.85	1
12	4	3	2	8190	128	6.84e-03	5.91e-02	2.19e-02	8.6	7	3.14	2
12	4	3	3	8190	128	6.15e-02	2.24e-01	1.33e-01	3.6	26	3.08	3
12	6	2	2	8190	128	2.93e-03	4.86e-02	1.78e-02	16.6	6	2.85	2
12	6	2	3	8190	128	1.03e-02	1.02e-01	4.15e-02	10.0	13	2.95	2
12	6	2	4	8190	128	5.42e-02	3.37e-01	1.21e-01	6.2	41	3.08	2
14	2	7	2	32766	256	9.92e-01	9.99e-01	9.96e-01	1.0	254	3.40	1
14	7	2	4	32766	256	3.08e-02	2.90e-01	1.00e-01	9.4	70	3.45	5
16	2	8	2	131070	512	9.96e-01	1.00e+00	9.97e-01	1.0	510	4.00	1
16	4	4	2	131070	512	1.83e-03	3.19e-02	8.21e-03	17.4	15	4.13	3
16	4	4	3	131070	512	3.11e-02	2.29e-01	8.28e-02	7.4	111	4.00	4
16	8	2	2	131070	512	9.16e-05	6.45e-03	2.21e-04	70.4	3	3.75	1
16	8	2	3	131070	512	4.58e-04	1.47e-02	1.93e-03	32.2	7	4.04	2
16	8	2	4	131070	512	1.92e-03	3.32e-02	8.28e-03	17.3	16	4.13	3
18	2	9	2	524286	1024	9.98e-01	1.00e+00	9.99e-01	1.0	1022	4.44	1
18	6	3	2	524286	1024	2.14e-04	1.62e-02	1.12e-03	76.0	14	4.55	2
18	6	3	3	524286	1024	1.17e-03	3.67e-02	5.51e-03	31.3	35	4.47	3
18	6	3	4	524286	1024	1.46e-02	4.01e-01	8.53e-02	27.4	346	4.58	14
18	9	2	3	524286	1024	2.29e-04	1.82e-02	1.13e-03	79.7	16	4.55	2
18	9	2	4	524286	1024	2.29e-04	1.82e-02	1.13e-03	79.7	16	4.55	2
20	2	10	2	2097150	2048	9.99e-01	1.00e+00	9.99e-01	1.0	2046	4.95	1
20	4	5	2	2097150	2048	4.73e-04	1.65e-02	1.68e-03	35.0	31	5.07	3
20	4	5	3	2097150	2048	1.56e-02	2.36e-01	6.32e-02	15.1	462	4.98	13
20	10	2	2	2097150	2048	1.14e-05	2.88e-03	1.24e-05	251.9	6	5.09	1
20	10	2	3	2097150	2048	1.14e-05	2.88e-03	1.24e-05	251.9	6	5.09	1
20	10	2	4	2097150	2048	1.14e-05	2.88e-03	1.24e-05	251.9	6	5.09	1
21	7	3	4	4194302	2896	3.90e-03	2.81e-01	1.25e-01	72.0	644	5.32	112
22	2	11	2	8388606	4096	1.00e+00	1.00e+00	1.00e+00	1.0	4094	5.46	1
24	2	12	2	33554430	8192	1.00e+00	1.00e+00	n/a	1.0	8190	5.95	n/a
24	4	6	2	33554430	8192	1.20e-04	8.35e-03	n/a	69.5	63	6.06	n/a
24	4	6	3	33554430	8192	7.81e-03	2.46e-01	n/a	31.6	1969	5.97	n/a
24	6	4	2	33554430	8192	1.43e-05	4.47e-03	n/a	312.5	30	6.06	n/a
24	6	4	3	33554430	8192	1.36e-04	1.25e-02	n/a	91.7	91	6.06	n/a
24	6	4	4	33554430	8192	3.78e-03	5.21e-01	n/a	137.7	3699	6.06	n/a
24	8	3	2	33554430	8192	1.67e-06	8.63e-04	n/a	517.2	7	6.08	n/a
24	8	3	3	33554430	8192	1.50e-05	3.46e-03	n/a	230.6	26	6.10	n/a
24	8	3	4	33554430	8192	1.22e-04	8.39e-03	n/a	68.9	64	6.06	n/a
24	12	2	2	33554430	8192	3.58e-07	5.52e-04	n/a	1543.2	3	6.30	n/a
24	12	2	3	33554430	8192	3.58e-06	2.12e-03	n/a	591.9	14	6.30	n/a
24	12	2	4	33554430	8192	2.04e-05	7.33e-03	n/a	359.5	50	6.06	n/a

Observations. In every row with $n \ge 4$ the class closure exceeds the direct weak set, by a factor of 1.8 to 1543. For fixed (n, M) the factor grows with N: for $n = 4, M = 2$ it is 8.6, 17.4, 35.0 and 69.5 at $N = 12, 16, 20, 24$. The 3-isogeny/Frobenius walk reaches part of the class closure. At $N = 20, n = 4, M = 3$ it is 6.32e-2 against 2.36e-1; at $N = 20, n = 10$ it is 1.24e-5 against 2.88e-3. The longest walk recorded is 112 steps ($N = 21, n = 7, M = 4$).

Derived. By Tate and MMT Thm 5(ii), weak curves number roughly $q^{M/n}$ while classes number about $4\sqrt{q}$. Once the weak count passes about $\sqrt{q}$, a constant share of classes holds a weak member. This is a counting statement, not an attack cost.

Not established. The cost of the isogeny beyond 3-isogeny steps; GHS genus, Jacobian order and subgroup transfer; any comparison with rho; transfer to cryptographic N.

4. Lead 2: Smith kernels per curve and per class

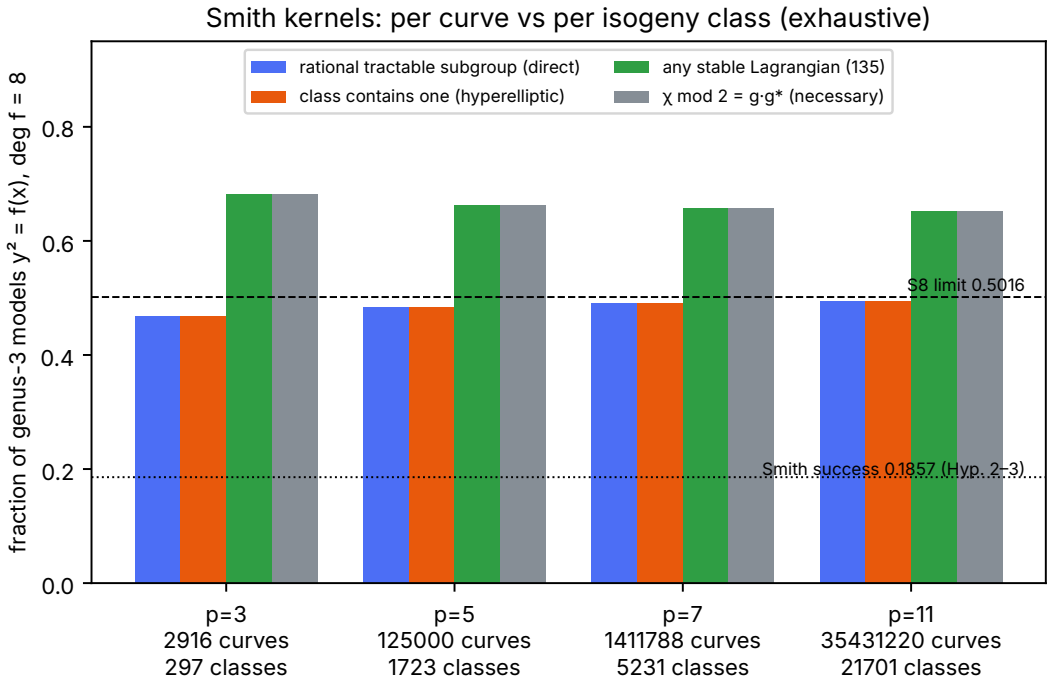


Figure 3: Exhaustive genus-3 models $y^2 = f(x)$, $\deg f = 8$, curve and twist. Dashed: the S_8 limit 0.50159. Dotted: Smith's success rate 0.1857.

p	curves	classes	direct tractable	class closure	any Lagrangian	$\bar{\chi} = g{\cdot}g^*$	charpoly mismatches	violations
3	2916	297	0.4678	0.4678	0.6818	0.6818	0	0
5	125000	1723	0.4840	0.4840	0.6626	0.6626	0	0
7	1411788	5231	0.4897	0.4897	0.6563	0.6563	0	0
11	35431220	21701	0.4943	0.4943	0.6514	0.6514	0	0

Observations. $\#S(H)$ for all 22 cycle types matches Smith's Lemma 1. $\sum_T P(T)\big(1-(3/4)^{\#S(H)}\big) = 0.185738$ reproduces Smith's 18.57% (eq. (12)), where $3/4$ comes from his Hypotheses 2–3. Without those hypotheses, the rational-tractable fraction is 0.50159. On every curve, $\chi_\pi \bmod 2$ equals the characteristic polynomial of the root permutation on $J[2]$ (0 mismatches). **The class closure equals the direct fraction for every p.**

Derived. On S_8, $\chi \bmod 2$ determines whether a tractable subgroup exists (finite check, table below). It is an isogeny invariant, so a walk inside the hyperelliptic locus never creates a tractable subgroup. It can only flip Smith's square conditions (Props. 2, 4); under his hypotheses this bounds the gain at 18.57% ⟶ 50.16%. That is open (`problems/smith-kernel-isogeny-class-walk.md`).

cycle type	#S_8	#S(H)	stable Lagrangians	$\bar{\chi}$ mod 2 (coeff. bits, palindromic)
1+1+3+3	1120	3	9	1000001
1+1+6	3360	1	1	1000001
2+3+3	1120	3	3	1000001
2+6	3360	1	3	1000001
1+1+1+1+1+1+1+1	1	105	135	1010101
1+1+1+1+1+1+2	28	15	15	1010101
1+1+1+1+2+2	210	9	15	1010101
1+1+1+1+4	420	3	3	1010101
1+1+2+2+2	420	7	7	1010101
1+1+2+4	2520	1	3	1010101
2+2+2+2	105	25	39	1010101
2+2+4	1260	3	3	1010101
4+4	1260	5	11	1010101
8	5040	1	1	1010101
3+5	2688	0	0	1011101
1+1+1+5	1344	0	0	1100011
1+2+5	4032	0	0	1100011
1+1+1+1+1+3	112	0	0	1110111
1+1+1+2+3	1120	0	0	1110111
1+2+2+3	1680	0	0	1110111

1+3+4	3360	0	0	1110111
1+7	5760	0	2	1111111

S_8 pair 0.50159, any 0.64444, S7 pair 0.43492

5. Where the leads meet

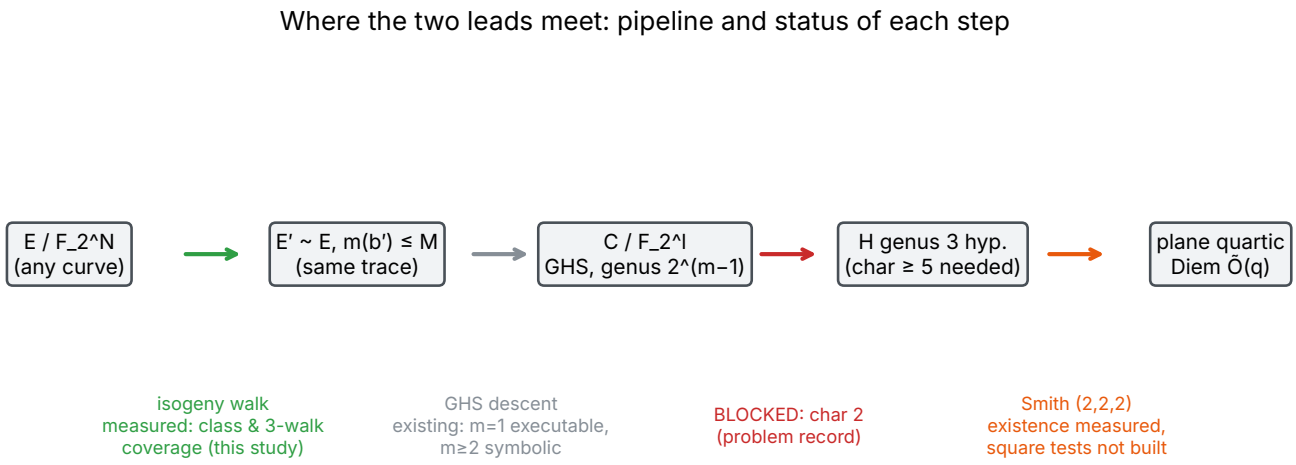


Figure 4: Status of each step of the proposed pipeline.

The pipeline does not compose as stated. GHS lands over $\mathbb{F}_{2^l}$, and Smith's trigonal construction "does not apply in characteristics 2 and 3" (Smith 2009). A derived reason: an ordinary genus-3 Jacobian in characteristic 2 has $\#J[2]\left(\overline{k}\right) = 2^3$. See `problems/ghs-to-smith-pipeline-characteristic-mismatch.md`. The end-to-end proportion is not computable from this study.

6. References

Maurer, Menezes, Teske, LMS JCM 5 (2002) 127–174. Gaudry, Hess, Smart, J. Cryptology 15 (2002). Smith, J. Cryptology 22 (2009), arXiv:0806.2995. Diem, ANTS 2006; Gaudry–Thomé–Thériault–Diem, Math. Comp. 2007 (cited costs, not re-derived). Hess, LMS JCM 7 (2004), not read in this round.